Aaron S. Martin

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Summary

Embedded software engineer with 4 years developing prototype vehicle-control software — electric-vehicle controllers, brake controllers, and vehicle control units — in embedded C and MATLAB/Simulink with generated C. Promoted to Senior Engineer in 2025; U.S. Citizen, clearable for defense work.

Skills

Embedded & Controls: Simulink, generated C, state-machine modeling, diagnostics, calibrations, CAN, SAE J1939, UDS
Verification & Tools: Python regression automation, HIL, SIL, CANalyzer, INCA, Git, Jira, Polarion, requirements traceability
Languages: C, Python, MATLAB, VHDL

Experience

Senior Engineer, International LLC (Navistar / Traton Group) – Lisle, IL Mar 2025 – Present

- Developed real-time brake, parking-brake, hold-function, and trailer-detection control logic for prototype/test-fleet vehicle control units in embedded C, including switch handling, startup behavior, and vehicle-state interlocks
- Delivered 8 peer-reviewed GitLab merge requests backed by requirements, CAN and runtime-data configuration, and tests — resolving 190/190 code-review discussion threads with zero unresolved at merge
- Authored or materially updated ~35 automated Python regression methods (32 in 5 new test files), expanding coverage across brake, hold-function, parking-brake, and trailer-detection behavior
- Owned 48 tracked Jira work items spanning controls, diagnostics, verification, and requirements — 39 completed, 39 rated high or critical priority
- Built local tooling and review practices for requirement-test traceability, test tagging, and CAN troubleshooting

Engineer, International LLC (Navistar / Traton Group) – Lisle, IL June 2022 – Mar 2025

- Built and maintained model-based electric-vehicle controller features in MATLAB/Simulink with generated C, runtime data interfaces, and calibrations for prototype/test-fleet software
- Implemented charging-status and display signaling logic, handling plug, charger, contactor, and vehicle-state edge cases
- Redesigned high-voltage contactor diagnostics (stuck-open and stuck-closed detection) with calibratable timing and threshold logic and exposed measurement points for validation
- Contributed to 527 Polarion requirements records for prototype electric-vehicle controllers, spanning control logic, diagnostics, signal interfaces, calibration, and power and energy behavior
- Supported 2 prototype-vehicle test events (charging-issue and recurring calibration support), troubleshooting networked thermal and auxiliary subsystems with CANalyzer and INCA

Projects

Simple GPU on an FPGA

- Implemented the OpenGL driver (C), a GLSL-to-custom-ISA shader translator (Lua), vertex-shader ISA extensions, and hardware performance counters (VHDL) for a 4-student soft GPU rendering on an Artix-7 FPGA (Xilinx Vivado)

Powder-Bed Metal 3D Printer — Senior Design, Team Lead

- Led a 6-person team rewriting the printer-control software in MATLAB — G-code compiled to auditable command sequences driving steppers, a function generator, and a Class-4 fiber laser with response validation

Education

Iowa State University – B.S. in Computer Engineering Dec 2021
Advanced coursework: Hardware Design for Machine Learning, GPU Architecture, Reconfigurable Hardware (graduate), CMOS Integrated Electronics (Cadence), Operating Systems